

Big Data & Apache Spark

62 MCQs with Answers — Assessment / Bifurcation Preparation

Based on the uploaded Big Data & Apache Spark preparation notes. The questions cover Big Data fundamentals, Hadoop, Spark architecture, execution model, RDDs, DataFrames, optimization, partitions, Spark UI, and Structured Streaming.

1. Which of the following is NOT one of the classic 4Vs of Big Data?

- A. Volume
- B. Velocity
- C. Variety
- D. Value

Answer: D. Value

2. What does "Volume" in Big Data refer to?

- A. Speed of data generation
- B. Scale/amount of data
- C. Data quality
- D. Data format

Answer: B. Scale/amount of data

3. Which type of data generally has no fixed schema?

- A. Structured data
- B. Relational data
- C. Unstructured data
- D. Tabular data

Answer: C. Unstructured data

4. Which is an example of unstructured data?

- A. SQL table
- B. Excel table
- C. Images and videos
- D. Relational database

Answer: C. Images and videos

5. Traditional RDBMS generally scale:

- A. Horizontally
- B. Vertically
- C. Randomly
- D. Through partitioning only

Answer: B. Vertically

6. Big Data frameworks primarily use:

- A. Vertical scaling
- B. Horizontal scaling
- C. Manual scaling
- D. Single-machine scaling

Answer: B. Horizontal scaling

7. Which is a major limitation of traditional RDBMS for Big Data?

- A. Lack of SQL
- B. Lack of horizontal scalability
- C. Lack of tables
- D. Lack of indexes

Answer: B. Lack of horizontal scalability

8. In distributed computing, data and computation are:

- A. Stored only on one machine
- B. Split across multiple nodes
- C. Always stored in RAM
- D. Stored only in databases

Answer: B. Split across multiple nodes

9. Parallel processing mainly helps to:

- A. Reduce throughput
- B. Increase processing time
- C. Increase throughput and reduce processing time
- D. Remove all failures

Answer: C. Increase throughput and reduce processing time

10. What does fault tolerance mean?

- A. Preventing all failures
- B. Ensuring failures do not stop the job
- C. Increasing CPU speed
- D. Removing partitions

Answer: B. Ensuring failures do not stop the job

11. What is HDFS?

- A. Resource manager
- B. Distributed file system
- C. Programming language
- D. Database

Answer: B. Distributed file system

12. HDFS stores data:

- A. Only on the NameNode
- B. In blocks across nodes
- C. Only in memory
- D. Only in a relational database

Answer: B. In blocks across nodes

13. Which Hadoop component stores metadata?

- A. DataNode
- B. NameNode
- C. YARN
- D. MapReduce

Answer: B. NameNode

14. Which Hadoop component stores actual data blocks?

- A. NameNode
- B. DataNode
- C. ResourceManager
- D. Driver

Answer: B. DataNode

15. MapReduce is primarily a:

- A. Memory-based streaming model
- B. Disk-based batch processing model
- C. Database system
- D. Visualization system

Answer: B. Disk-based batch processing model

16. What is the primary role of YARN?

- A. Store data blocks
- B. Manage cluster resources and schedule applications
- C. Execute SQL queries
- D. Create DataFrames

Answer: B. Manage cluster resources and schedule applications

17. Hadoop detects worker-node failures using:

- A. Logs
- B. Heartbeat signals
- C. SQL queries
- D. DataFrames

Answer: B. Heartbeat signals

18. Apache Spark is a:

- A. Database
- B. File system
- C. Distributed data processing engine
- D. Visualization tool

Answer: C. Distributed data processing engine

19. Which of the following is NOT a Spark capability?

- A. Batch processing
- B. Streaming
- C. Machine learning
- D. File system storage

Answer: D. File system storage

20. Spark is generally faster than MapReduce because of:

- A. Larger databases
- B. In-memory computation
- C. SQL only
- D. Vertical scaling

Answer: B. In-memory computation

21. Which Spark component controls execution and builds the DAG?

- A. Executor
- B. Driver
- C. DataNode
- D. Cluster Manager

Answer: B. Driver

22. Executors are responsible for:

- A. Allocating cluster resources
- B. Running tasks and storing cached data
- C. Building SQL queries only
- D. Managing NameNode metadata

Answer: B. Running tasks and storing cached data

23. Which component allocates cluster resources?

- A. Driver
- B. Executor
- C. Cluster Manager
- D. DataFrame

Answer: C. Cluster Manager

24. Which of the following can be a Spark Cluster Manager?

- A. YARN

- B. Standalone
- C. Kubernetes
- D. All of the above

Answer: D. All of the above

25. Scheduling logic in Spark primarily belongs to:

- A. Executor
- B. Driver
- C. DataNode
- D. HDFS

Answer: B. Driver

26. Spark uses:

- A. Eager evaluation only
- B. Lazy evaluation
- C. Manual evaluation
- D. Random evaluation

Answer: B. Lazy evaluation

27. When is a Spark transformation actually executed?

- A. Immediately
- B. When the application starts
- C. When an action triggers execution
- D. When DataFrame is created

Answer: C. When an action triggers execution

28. What is the correct Spark execution flow?

- A. User Code → Tasks → DAG → Stages → Executors
- B. User Code → DAG → Stages → Tasks → Executors
- C. DAG → User Code → Executors → Tasks
- D. Executors → DAG → Stages → User Code

Answer: B. User Code → DAG → Stages → Tasks → Executors

29. Which operation generally causes a shuffle?

- A. filter
- B. map
- C. groupBy
- D. select

Answer: C. groupBy

30. Wide transformations can cause:

- A. Shuffle and stage boundaries
- B. Database creation

- C. File deletion
- D. Driver termination

Answer: A. Shuffle and stage boundaries

31. RDD stands for:

- A. Random Distributed Database
- B. Resilient Distributed Dataset
- C. Relational Distributed Data
- D. Restructured Data Definition

Answer: B. Resilient Distributed Dataset

32. RDDs are:

- A. Mutable and centralized
- B. Immutable and distributed
- C. Mutable and centralized
- D. Only stored on the Driver

Answer: B. Immutable and distributed

33. RDD fault tolerance is primarily achieved through:

- A. Indexing
- B. Lineage
- C. SQL
- D. Caching only

Answer: B. Lineage

34. If an RDD partition is lost, Spark can:

- A. Stop permanently
- B. Recompute it using transformation history
- C. Delete the complete RDD
- D. Restart the database

Answer: B. Recompute it using transformation history

35. Which of the following is a transformation?

- A. count()
- B. collect()
- C. first()
- D. filter()

Answer: D. filter()

36. Which of the following is an action?

- A. map()
- B. filter()
- C. select()

D. count()

Answer: D. count()

37. What happens when only transformations are defined and no action is called?

- A. Transformations execute immediately
- B. Nothing is executed
- C. Spark automatically calls collect()
- D. Driver stops

Answer: B. Nothing is executed

38. Which operation triggers Spark execution?

- A. filter()
- B. map()
- C. count()
- D. select()

Answer: C. count()

39. DataFrames provide:

- A. A lower-level API than RDD
- B. A higher-level API over RDDs
- C. A replacement for HDFS
- D. A cluster manager

Answer: B. A higher-level API over RDDs

40. What is the modern unified entry point for Spark?

- A. SparkContext only
- B. SparkSession
- C. Executor
- D. DataNode

Answer: B. SparkSession

41. spark.sql() returns:

- A. RDD
- B. Python list
- C. Spark DataFrame
- D. String

Answer: C. Spark DataFrame

42. Which format is columnar and optimized for analytics?

- A. CSV
- B. TXT
- C. Parquet
- D. XML

Answer: C. Parquet

43. What does inferSchema=True do when reading CSV?

- A. Removes headers
- B. Automatically detects data types
- C. Converts CSV to JSON
- D. Creates partitions only

Answer: B. Automatically detects data types

44. JDBC data can be read using:

- A. spark.read.jdbc()
- B. spark.jdbc.read()
- C. spark.sql.jdbc()
- D. DataFrame.jdbc()

Answer: A. spark.read.jdbc()

45. Which optimizer improves Spark SQL/DataFrame execution plans?

- A. JVM Optimizer
- B. Catalyst Optimizer
- C. Hadoop Optimizer
- D. Python Optimizer

Answer: B. Catalyst Optimizer

46. Which is generally faster?

- A. Python UDF
- B. Built-in Spark function
- C. Both always have identical performance
- D. Python loop

Answer: B. Built-in Spark function

47. Python UDFs can introduce:

- A. Python-JVM overhead
- B. HDFS metadata
- C. NameNode failure
- D. SQL syntax errors only

Answer: A. Python-JVM overhead

48. Which method can be used to inspect a Spark query execution plan?

- A. df.show()
- B. df.explain()
- C. df.print()
- D. df.plan()

Answer: B. df.explain()

49. The number of Spark tasks depends mainly on:

- A. Number of columns
- B. Number of partitions
- C. Number of databases
- D. Number of DataFrames

Answer: B. Number of partitions

50. Which operation increases parallelism but causes a shuffle?

- A. coalesce()
- B. repartition()
- C. filter()
- D. select()

Answer: B. repartition()

51. Which operation generally reduces partitions with minimal shuffle?

- A. repartition()
- B. coalesce()
- C. groupBy()
- D. join()

Answer: B. coalesce()

52. Too few tasks while processing Big Data usually indicates:

- A. Too many partitions
- B. Insufficient partitions
- C. Too many executors
- D. Too much caching

Answer: B. Insufficient partitions

53. Which Spark UI tab provides stage and task-level execution details?

- A. Executors
- B. Stages
- C. Storage
- D. Environment

Answer: B. Stages

54. Which Spark UI tab shows executor memory and task summaries?

- A. Stages
- B. SQL
- C. Executors
- D. Jobs

Answer: C. Executors

55. Spark Streaming is mainly used for:

- A. Near real-time processing
- B. Only batch processing
- C. Database administration
- D. File compression

Answer: A. Near real-time processing

56. Structured Streaming can use:

- A. Only batch execution
- B. Micro-batch or continuous execution
- C. Only MapReduce
- D. Only RDD execution

Answer: B. Micro-batch or continuous execution

57. Which function is used for rolling/sliding aggregations?

- A. groupBy()
- B. window()
- C. partition()
- D. slide()

Answer: B. window()

58. The window slide duration must be:

- A. Greater than batch interval
- B. Less than batch interval
- C. A multiple of the batch interval
- D. Unrelated to batch interval

Answer: C. A multiple of the batch interval

59. Checkpointing is especially important for:

- A. Stateless operations only
- B. Stateful operations
- C. CSV reading
- D. DataFrame creation

Answer: B. Stateful operations

60. Which function controls how late events are handled?

- A. withWatermark()
- B. withCheckpoint()
- C. lateData()
- D. watermarkOnly()

Answer: A. withWatermark()

61. Backpressure allows Spark to:

- A. Delete old data

- B. Adapt ingestion rate when sources are fast
- C. Increase database size
- D. Remove partitions

Answer: B. Adapt ingestion rate when sources are fast

62. Which is a common streaming source mentioned in the notes?

- A. Kafka
- B. MySQL
- C. Excel
- D. HTML

Answer: A. Kafka

Important Bifurcation Traps

If the question says...	Think...
Recomputation after failure	Lineage
groupBy() makes job slow	Shuffle
Nothing is executing	Missing Action
Streaming restarts safely	Checkpointing
Metadata	NameNode
Actual HDFS blocks	DataNode
Resource management	YARN / Cluster Manager
Scheduling logic	Driver
Running tasks	Executors
Increase partitions	repartition()
Reduce partitions	coalesce()
SQL/DataFrame optimization	Catalyst
Late streaming data	withWatermark()
Analytics columnar format	Parquet

Quick Revision: 4Vs = Volume, Velocity, Variety, Veracity. Driver = scheduling/control. Executors = tasks/cache. NameNode = metadata. DataNode = data blocks. Transformations are lazy; actions trigger execution. groupBy/join can cause shuffle. Lineage supports RDD recovery. Checkpointing supports streaming state recovery.